



Sustainable biorefinery of *Ulva stenophylloides*, a green-tide species from the South Pacific coast: A study on antioxidant, oligosaccharide, and ulvan extraction

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ABSTRACT

In this study, biomass characterization and sustainable biorefinery were determined using microwave-assisted extraction to recover valuable biocompounds from the green-tide species *Ulva stenophylloides*. The biomass exhibits concentrations of 21 % for proteins; 24.6 % for sulfate; 2.1 % for lipids; 27 % for carbohydrates; 15 % and 11 % for chlorophylls *a* and *b*, respectively; and 4.6 % for carotenoids. The temperature of the microwave treatment influenced both the yield and quality of biocompounds, with higher concentrations of phenolic compounds and antioxidant activities observed in the soluble extracts at temperatures between 180 °C and 220 °C, and higher concentrations of oligosaccharides obtained at temperatures from 120 °C to 200 °C. Microwave treatment at 120 °C yielded the highest content of ulvans (21 %), as well as favorable molecular weight and viscoelastic properties. Furthermore, the ulvan extract demonstrated no toxic effect in both *in vitro* and *in vivo* toxicological assays. These findings highlight the potential of green tides as a renewable biomass source for various biocompounds, opening up possibilities for future biotechnology research.

1. Introduction

The proliferation of *Ulva* species has become a more common and widespread coastal phenomenon worldwide, possibly due to anthropogenically altered ocean conditions, thereby contributing to global climate change [1,2]. In fact, along the central Chilean coast, there have been blooms of these algae that persist almost year-round over the past 20 years, leading to environmental, economic, and social issues that affect local government and communities (Municipal Report,

Algarrobo). In this context, the Municipality of Algarrobo has reported approximately 400 t of dry biomass extracted from sandy beaches in the last 4 years. Particularly, we recently characterized these green tides (from molecular, morphological, and ecological perspectives), marking it the first study of this kind in the country and the South Pacific [3]. We concluded that the predominant species in the recorded green tides is *Ulva stenophylloides* L.G. Kraft, Kraft & R.F. Waller, covering an average of 53 % of the coastal zone across all seasons and intertidal levels. This species had previously only been described in Australia, New Zealand,

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and Southern Africa [4]. There are no other studies in Chile or in the region characterizing green tides, preventing comparisons with other affected areas in the country. These green tides grow and detach from hard substrates, accumulating large biomass while suspended in the water column [3]. Similar effects have been observed in coastal zones worldwide, highlighting the global scale of this anthropogenic issue [4]. In this context, it is worth considering the potential use of the biomass of *Ulva* species from green tides for the revalorization of biomolecules of interest, among other benefits. Indeed, biomass production can provide benefits as a source of food and feed for industries such as nutraceuticals and cosmetics, owing to the presence of bioactive metabolites (primary and secondary) [5]. Therefore, its utilization at a productive level in the search for biocompounds could allow for large-scale management.

The authors of [6] recently summarized some of the interesting properties of the main components and fractions of algae belonging to the genus *Ulva*, along with examples of their main applications. Various studies have indicated that *Ulva* biomass has a high carbohydrate content (37–79 %), high levels of proteins (7–32 %), and diverse lipids (0.4–2 %; for example, ω -3 and ω -6 fatty acids, eicosapentaenoic acid, docosapentaenoic acid, and docosahexaenoic acid), emphasizing the high potential of this genus in producing diverse biocompounds with significant relevance to human and animal health, e.g., [7–14]. However, the literature also indicates that the types of compounds and their concentration and yield depend on various factors such as species, culture type, biomass collection season, coastal area, and other environmental factors. These characteristics also influence the type of application, including their use as novel materials such as sustainable thermal insulation materials to reduce the energy demand of buildings [15,16], bioremediation [17,18], fungicides [19,20], and cosmetic applications [21,22], among others.

The biorefinery of algal biomass to produce value-added products for food, feed, chemicals, materials, and energy (in the form of biofuels, power, and/or heat) through innovative and sustainable bioprocess systems aligns with several of the Sustainable Development Goals (SDGs) [23]. Among such novel techniques, intensification by microwave heating shows promise, mainly due to the reduction in operation time and energy consumption. Careful selection of the conditions is required to tune the product distribution and properties, as depolymerization of polysaccharides could occur. Biomass from green tides can serve as a valuable local source with a lower cost, making the process economically viable. Therefore, biorefinery techniques will need to be adapted to the species of each area and their composition [24]. Research on different biorefinery approaches for the valorization of green seaweeds has been reported in recent years, including studies to obtain ulvan, protein, and biogas [25]; salt, starch, lipids, ulvan, protein, and cellulose [26]; protein, sugar, and phenolics [27]; or feed, chemicals, and fuels [28]. However, most of the techniques used are based on acid, alkaline, or enzymatic extraction methods [25,29,30]. In this context, our central objective was to carry out sustainable biorefinery of green tides from Chile, aiming to revalorize their biomass through a simple and sustainable method to obtain bioproducts. For example, ulvans can be extracted via microwave-assisted extraction methods using water as the solvent, and the residue from this extraction process can then be processed to extract compounds of interest for various industries, such as amino acids, antioxidants, and sugars. Ulvans are polysaccharides found in the cell walls of *Ulva* species, primarily composed of sulfated rhamnose, uronic acids (D-glucuronic and L-iduronic), and xylose [31]. The structure of ulvans is unique with the presence of sulfated L-rhamnose and iduronic acid and contains two main repeating disaccharides, ulvanobiuronic acids (types A and B) and ulvanobioses (type U) [31]. Ulvans have a wide range of applications, including in biomaterials (such as wound dressings, tissue engineering, and biofilm prevention), nutraceuticals (with antiviral, antioxidant, anti-hyperlipidemic, anticancer, and immunostimulant properties), functional foods, and agriculture [31,32]. The biological activities of ulvans are closely linked to their chemical structure, with *Ulva* species as a

biomass source. Ulvans are part of the polysaccharide market, which was valued at USD 1.9 billion in 2022 and is forecasted to reach USD 3.1 billion by 2030, showing steady growth in recent years and projected to continue growing at a CAGR of 7.5 % between 2023 and 2030 (source: <https://www.credenceresearch.com/report/algal-polysaccharides-market>).

While the global macroalgae market was estimated to reach USD 19 billion by 2024, a World Bank report [33] indicates the emergence of 10 global algae markets with a potential growth projection of up to USD 11.8 billion by 2030. These markets primarily encompass nutraceuticals, biostimulants, and the feed sector. The report also underscores the potential of biorefinery as an economically viable process, allowing for the derivation of multiple products from a single seaweed source. In this context, it is intriguing to explore products that can be obtained from green tides through biorefinery. Therefore, in this study, we present an efficient biorefinery process for obtaining diverse bioactive fractions and compounds with significant implications for human and animal health from the Chilean green-tide species *Ulva stenophylloides*, including the ulvan fraction. Our findings lay the groundwork for future management efforts in green algae aquaculture and biotechnological applications. We provide basic information for new applications in the country and region regarding the use of this type of biomass.

2. Materials and methods

2.1. Sampling collection and raw material treatment

The sampling of *Ulva stenophylloides* was conducted in the summer of 2023 during low tide along 150 m of the rocky intertidal zone of Los Tubos Beach in Algarrobo Bay, located south of Valparaíso, Chile (33°21.891' S, 71° 40.763' W). This site is characterized by a sedimentary rocky platform, with channels and heterogeneity formed mainly by granitic rock boulders [3]. Four kilograms of the collected biomass was immediately stored in plastic bags containing seawater and transported to the laboratory (www.lebma.cl) in a cooler at 5–7 °C. Later, the raw material was rinsed according to the protocol described in [34], using tap water and 0.22 μ m filtered seawater. Subsequently, the biomass was dried at 28–30 °C using a convective air oven (MLab Scientific, MDO-065FB) until a constant weight was achieved (72 h). Afterwards, the dried biomass was micronized to an average particle size of less than 300 μ m using a grinder. The raw material was stored under vacuum and kept at –20 °C before the chemical characterization and extraction process.

2.2. Characterization of dried raw material

2.2.1. Moisture, ash, and sulfate contents

The moisture content of the *Ulva stenophylloides* biomass stored at –20 °C was determined gravimetrically. Samples (approximately 1 g, $n = 3$) were placed in glass tubes and dried in a convective air oven at 105 °C for 48 h until a constant weight was achieved. The ash content ($n = 3$) was also determined gravimetrically using a muffle furnace at 575 °C for 6 h. The corresponding sulfate content ($n = 3$) was analyzed following the method developed by Gómez-Ordóñez, Alonso, and Rupérez [35]. This characterization was performed via ion chromatography (Metrohm Advanced IC-861, Switzerland) with a Metrosep A Supp 5–250 column (250 $\times$ 4 mm). The mobile phase consisted of sodium carbonate/sodium bicarbonate at concentrations of 3.2 mM and 1 mM, respectively, with a flow rate of 0.70 mL/min. The IC-819 detector was employed.

2.2.2. Carbohydrate content

To characterize the carbohydrates present in the raw material, two-step acid hydrolysis was performed in triplicate, following an established protocol (NREL/TP-510-42618) [36]. In the first step, polysaccharides were hydrolyzed into oligosaccharides by adding 5 mL of 72

% (w/w) H₂SO₄ to a test tube containing 0.5 g of the raw material. The mixture was heated at 30 °C for 1 h in a water bath with regular stirring. Then, the content was transferred to a Pyrex flask. The acid concentration was adjusted to 4 % (w/w) by adding distilled water, and the system was autoclaved for 1 h at 121 °C to break down oligosaccharides into monosaccharides. The system was then filtered using a Gooch filter, with pore size n°3. The filter was dried in an oven at 105 °C to determine the acid-insoluble residue (AIR), while the liquid phase was filtered (0.45 µm filters) and analyzed via high-performance liquid chromatography in an Aminex HPX87H column (300 × 7.8 mm, BioRad, USA) operating at 60 °C, with 0.003 M sulfuric acid (Sigma-Aldrich, USA) as the mobile phase (0.6 mL/min). The standards employed were dextrans with a molecular weight from 1000 to 80.000 g/mol (Fluka, USA).

After integration of the chromatograms from the samples and standards, the concentration of carbohydrates in the raw material was calculated by summing the concentration of each monosaccharide in the sample using the following formula:

$$C_{\text{monosaccharide}} \left(\frac{\text{g}}{100 \text{ g raw material}} \right) = F \times \frac{A_{\text{sample}} \times [S]}{A_{\text{standard}}} \times \frac{W_{LP}}{1000 \times F_a} \times \frac{1}{W_{\text{sample}} \times (1 - H)} \times 100 \quad (1)$$

where

F = Loss correction factor

$$= \frac{\text{Total solid and liquid phases' weight after the autoclave}}{\text{Total solid and liquid phases' weight before the autoclave}} \quad (2)$$

$$A_{\text{sample/Standard}} = \frac{\text{Samples' or standard's area of the monosaccharide of interest}}{\text{area of the monosaccharide of interest}} \quad (3)$$

$$[S] = \text{Monosaccharide's concentration in the standard in g/L} \quad (4)$$

$$W_{LP} = \text{Weight of the liquid phase of the partial dissolution in g} \\ = \text{Weight}_{\text{dissolution}} - \text{Weight}_{\text{solid sample}} \times (1 - \text{Moisture content}) \times \text{AIR} \quad (5)$$

$$\text{AIR} = \text{Acid - insoluble residue (g AIR/g raw material)} \quad (6)$$

$$F_a = \text{Anhydro factor} = \frac{\text{Monomer's molecular weight}}{\text{Monomer molecular weight when polysaccharide} (= \text{Monomer's MW} - \text{H}_2\text{O MW})} \quad (7)$$

$$W_{\text{sample}} = \text{Raw material's weight (g)} \quad (8)$$

$$H = \text{Humidity of the raw material (g H}_2\text{O/g raw material)} \quad (9)$$

2.2.3. C, N, H, protein, lipid, chlorophyll, and carotenoid contents

The carbon, hydrogen, and nitrogen contents were determined using an elemental analyzer (Thermo Flash EA 1112, Germany). The gas flow rates were set at 130 mL/min for helium, 100 mL/min for the reference, and 250 mL/min for oxygen. The analyzer operated at an oxidation and reduction furnace temperature of 900 and 680 °C, respectively. Multiple analyses were conducted using a 6 × 5 mm, 2.0 m column (Cromlab, Spain), with the column temperature set at 50 °C and the retention time at 420 s. Aspartic acid was used as a standard pattern [37]. Total protein content (n = 3) was estimated by means of the Kjeldahl method using a nitrogen conversion factor of 5.77, according to [38].

Total lipids of seaweeds (n = 3) were determined using a chloroform/methanol mixture (2:1 v/v), according to the Folch method [39]. The extraction was followed by transesterification to calculate the fatty acid content [40]. The extraction and determination of the chlorophyll content (a and b) (n = 3) were performed according to [41]. The results are expressed as mg/g dry weight of biomass as follows:

$$\text{Chlorophyll a} = \frac{[12.7 (A_{663}) - 2.69 (A_{645})] * \text{Vol. extraction}}{\text{weight of the sample}} \quad (10)$$

$$\text{Chlorophyll b} = \frac{[22.9 (A_{645}) - 4.68 (A_{663})] * \text{Vol. extraction}}{\text{weight of the sample}} \quad (11)$$

where (A₆₆₃) is the absorbance at 663 nm and (A₆₄₅) is the absorbance at 645 nm.

Finally, the extraction and determination of the carotenoid content were performed according to [41,42], using the same method used for chlorophyll estimation. The content is expressed as mg/g dry weight as follows:

$$\text{Carotenoids} = \frac{[4 (A_{480})] * \text{Vol. extraction}}{\text{weight of the sample}} \quad (12)$$

where (A₄₈₀) is the optical density at 480 nm and 4 is the correction factor.

2.3. Microwave-assisted hydrothermal treatment of dried ground biomass

Microwave-assisted extraction was conducted on the dried ground biomass using an Anton Paar Microwave Reactor Monowave 450 (Austria), which was equipped with an autosampler MAS 24. Microwave vials were filled with the ground biomass and distilled water at a solid-to-liquid ratio of 1:30 (w/w), following the protocols described in [37,43].

The extraction experiments were performed at different temperatures of 120 °C, 140 °C, 160 °C, 180 °C, 200 °C, and 220 °C, with three replicates each. The operation conditions (solid-liquid ratio and temperatures) were selected based on previous studies on green seaweeds [37]. In all experiments, the samples were heated at the specified temperature for 6 min, with a rotation speed of 40 × g and a power of 850 W, after which they were cooled to 50 °C.

In order to separate the liquid and solid phases, vacuum filtration was performed at 20 °C. The solid phases were dried at 30 °C in a

convective air oven, stored under vacuum, and kept at −20 °C. The liquid phases were utilized for ulvan extraction, determination of oligosaccharide and phenolic contents, and antioxidant analysis. The solid phases will be characterized in a subsequent study, as we are considering various potential uses.

2.4. Liquid phase analysis

2.4.1. Extraction and characterization of ulvan fraction

The liquid phases were precipitated with 95 % ethanol at a ratio of 1:1.5 (v/v) for 12 h at 4 °C. Ulvan fractions were obtained by centrifugation at 4000 × g for 15 min at 15 °C. The supernatants were collected and stored at −20 °C for further analysis. Ulvans were dried in an air oven at 30 °C and stored at room temperature (20–25 °C) until further structural characterization by means of high-performance size-exclusion chromatography (HPSEC), Fourier-transform infrared attenuated total

reflectance (FTIR-ATR), proton nuclear magnetic resonance (^1H NMR), and rheological analysis, as described in [37]. The HPSEC analysis was performed using a high-performance liquid chromatography equipment with an IR detector (Agilent). A TSKGel SuperMultipore PW-H (6×150 mm) column with a TSKGel guard column SuperMP (PW)-H (4.6×35 mm), both from Tosoh Bioscience (Germany), was used to evaluate the molar mass distribution. FTIR-ATR spectra were acquired using an infrared spectrophotometer (Nicolet 6700, Thermo Scientific, USA). The samples were mixed with KBr, pressed, and dehydrated using an infrared lamp for 30 min. FTIR-ATR profiles were recorded between 600 and 1800 nm at 30 scan/min employing the OPUS-2.52 software (Opus Software Limited, UK). ^1H NMR experiments were conducted employing a Bruker ARX400 spectrometer (Bruker BioSpin GmbH, Germany) operating at 400 MHz and using 3-(trimethylsilyl)-L-propane sulfonic acid (Sigma-Aldrich, USA) as the internal standard and deuterated water as the solvent.

Rheological measurements of the corresponding aqueous ulvan solutions (1 % w/w) were conducted using a controlled-stress rheometer (MCR302, Anton Paar, Germany). The samples were placed on a sand-blasted parallel plate (1 mm gap, 25 mm diameter), sealed with light paraffin oil, covered by solvent trap, and rested for 5 min before rheological testing. Steady-state shear viscosity ($1\text{--}100\text{ s}^{-1}$) and small-amplitude oscillatory shear ($1\text{--}10$ Hz, 5 Pa) measurements were performed at 25°C at least in duplicate.

2.4.2. Oligosaccharide content

The liquid phases obtained after ulvan precipitation were dialyzed for 72 h (Float-A-Lyzer®G2, USA). Subsequently, post-hydrolysis was conducted on the dialyzed soluble extracts to determine and quantify their oligosaccharide content [37]. For this, 5 g of each liquid phase was autoclaved for 20 min at 121°C with H_2SO_4 at 4 % (w/w). The resulting solution was then filtered using $0.45\text{ }\mu\text{m}$ filter and analyzed via high-performance liquid chromatography in an Aminex HPX87H column (300×7.8 mm, BioRad, USA) operating at 60°C , with 0.003 M sulfuric acid (Sigma-Aldrich, USA) as the mobile phase (0.6 mL/min).

The concentration of oligosaccharides in the extracts was calculated by summing the concentration of each monosaccharide using the following formula:

$$C_{\text{monosaccharide}} \left(\frac{\text{g}}{100\text{ g of extract}} \right) = \frac{F_{\text{Loss}}}{F_{\text{Dilution}}} \times \frac{[S] \times A_{\text{sample}}}{A_{\text{standard}}} \times \frac{1}{\text{DC}} \times 100 \quad (13)$$

where

$$F_{\text{Loss}} = \text{Correction factor of the loss due to heat} = \frac{\text{Weight of sample + acid before autoclave}}{\text{Weight of sample + acid after autoclave}} \quad (14)$$

$$F_{\text{Dilution}} = \text{Correction factor of the dilution} = \frac{\text{Weight sample}}{\text{Weight sample + acid}} \quad (15)$$

$$A_{\text{sample/Standard}} = \frac{\text{Samples's or standart's area of the monosaccharide of interest}}{\text{area of the monosaccharide of interest}} \quad (16)$$

$$[S] = \text{Monosaccharide's concentration in the standard} \left(\frac{\text{g}}{\text{L}} \right) \quad (17)$$

$$\text{DC} = \text{Dry content of the sample} \left(\frac{\text{g}}{\text{L}} \right) \quad (18)$$

2.4.3. Antioxidant analysis

2.4.3.1. Total phenolic compounds. Total phenolic compounds were spectrophotometrically determined at $\lambda = 765\text{ nm}$ using the Folin

method, as described in [44], with 0.25 mL of soluble extracts. A standard curve was prepared using gallic acid concentrations ranging from 0.009 g/L to 0.09 g/L.

2.4.3.2. Trolox equivalent antioxidant capacity. The Trolox equivalent antioxidant capacity (TEAC) was determined according to [45]. TEAC reagent was diluted with PBS until the mixture reached an absorbance of 0.7 ± 0.1 nm at 734 nm. The standard curve was established by adding 1 mL of the reagent to 10 μL of Trolox solutions at concentrations ranging from 2 mM to 0.10 mM. After stirring and incubation for 6 min in a water bath at 30°C , the absorbance was measured at $\lambda = 734\text{ nm}$. A control measure was conducted with distilled water. The same process was carried out using the soluble fractions. In all cases, 3 replicates were measured for each soluble fraction.

2.4.3.3. DPPH antioxidant assay. The scavenging effect of the extracts on DPPH (2,2-diphenyl-1-picrylhydrazyl) radicals was estimated using the spectrometric method proposed in [46]. For this, 1 mL of $6 \times 10^{-5}\text{ M}$ of a methanolic solution of DPPH was added to 25 μL of each soluble fraction. The reduction of the absorbance was measured at $\lambda = 515\text{ nm}$ during 16 min. In all cases, three replicates were measured for each soluble fraction.

The percentage of inhibition (PI) of each soluble fraction was determined as follows:

$$PI(\%) = \frac{[Abs(t = 0\text{ min}) - Abs(t = 16\text{ min})]}{Abs(t = 0\text{ min})} \times 100 \quad (19)$$

2.4.3.4. Ferric reducing antioxidant power (FRAP) assay. FRAP reagent was prepared using a previously described method [47]. In brief, a mixture of 300 mM acetate buffer (pH 3.6), 10 mM TPTZ (2,4,6-tripyridyl-s-triazine) solution in 40 mM HCl, and 20 mM $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$ was prepared in a 10:1:1 ratio. Then, 1.5 mL of the FRAP reagent was mixed with 50 μL of the liquid fraction, and the absorbance was measured at $\lambda = 593\text{ nm}$ after 6 min of incubation at room temperature. A calibration curve was prepared using 0.1 to 2 mM of ascorbic acid. In all cases, three replicates were measured for each soluble fraction.

2.5. Toxicological assays of ulvan extracts obtained at 120°C

Ulvan extracts have garnered significant interest in biomedical applications due to their diverse range of beneficial properties, including antioxidant, antitumor, anticoagulant, antiviral, anti-inflammatory, and immune-modulatory effects [31]. When considering natural products such as ulvans for biomedical use, assessing their cytotoxicity is crucial to further investigate their biological activities and potential therapeutic applications. Thus, *in vitro* and *in vivo* toxicological assays were conducted on the ulvan fractions obtained with a major yield at 120°C .

2.5.1. Toxicological MTT assay with neuro-2a cells

Neuro-2a cells (N2A, mouse neuroblastoma, ATCC-CCL-131) were maintained in DMEM medium supplement with a mix of 10 % bovine fetal serum (Hyclone cytiva #HC.SH30396.039), $1 \times$ GlutaMAX supplement (Gibco, #35050061), and $1 \times$ antibiotic/antimycotic (Gibco #3505006) in 15 cm culture dishes at 37°C in a humidified 5 % $\text{CO}_2/95\%$ incubator. The ulvan extract powder was dissolved in phosphate buffered saline (PBS) at a concentration of 1000 $\mu\text{g/mL}$ (medium containing serum induces the precipitation of the ulvan extract) and filtered (syringe filter of $0.22\text{ }\mu\text{m}$). N2A cells were seeded in 48-well plates at a concentration of 80,000 cells/well and treated 24 h later. The treatment was carried out in low-serum DMEM (DMEM with 1 % bovine fetal serum, $1 \times$ glutamate supplement, $1 \times$ antibiotic/antimycotic) with different concentrations of ulvan extract (10, 50, 100, 200, 500 $\mu\text{g/mL}$) for 16 h. Each set point had its own control which included PBS only.

The 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyl tetrazolium bromide (MTT) assay was performed as described previously, with minor

modifications [48]. The MTT salt (Sigma, # M5655) was dissolved in DMEM without phenol red at a concentration of 0.5 mg/mL and filtered (syringe filter of 0.22 μ m). After the cells were treated with different concentrations of the ulvan extract, 200 μ L of the MTT solutions was added to each well and incubation was allowed for 180 min at 37 °C. After this period, the medium was removed, and the crystal of formazan was dissolved in 200 μ L of dimethyl sulphoxide (DMSO) added to each well, with constant stirring for 30 min. The absorbance was measured using a microplate spectrophotometer (Synergy H1, Biotek, Winooski, Vermont, USA) at wavelengths of 570 and 620 nm. The difference between the two measurements was used to calculate the percentage of cell survival with respect to the control. The results were expressed as percentage of viability, considering the control as 100 %.

2.5.2. *Caenorhabditis elegans* studies: maintenance and experimental protocol

Nematodes were cultured under standard laboratory conditions at 20 °C on nematode growth medium (NGM) agar plates, seeded with *Escherichia coli* (OP50) as a food source, following established procedures [49]. The wild-type nematode strain N2 utilized in this study was obtained from the *Caenorhabditis* Genetics Center, which is supported by the NIH National Center for Research Resources (NCRR).

Ulvan extracts were administered to wild-type *C. elegans* individuals along with OP50. Dry ulvan extract was dissolved in ethanol and diluted with OP50 to achieve different concentrations of 0.24 mg/mL, 0.24 μ g/mL, and 0.4 ng/mL. The seeded plates were left to dry in the dark overnight in a laminar flow chamber and stored at 4 °C for no longer than 1 week before use.

To assess the potential toxic effects of the extracts on adult worms, age-synchronized one-day-old adult worms were exposed to different concentrations of ulvan extract. After 48 h, the viability of individuals and their normal growth were evaluated.

Fig. 1. shows the steps of the biorefinery and characterization analyses of *Ulva stenophylloides* biomass to recover diverse biomolecules with important commercial and biomedical potentials.

2.6. Statistical analysis

To determine the effect of temperature during microwave-assisted extraction on the antioxidant features of *Ulva stenophylloides*, such as FRAP assay values, TEAC, phenolic compound content, and percentage of inhibition (PI) of DPPH, Levene's test for homogeneity of variance and Shapiro–Wilk normality test were performed. Logarithmic transformation was applied to the PI data. Subsequently, Kruskal–Wallis

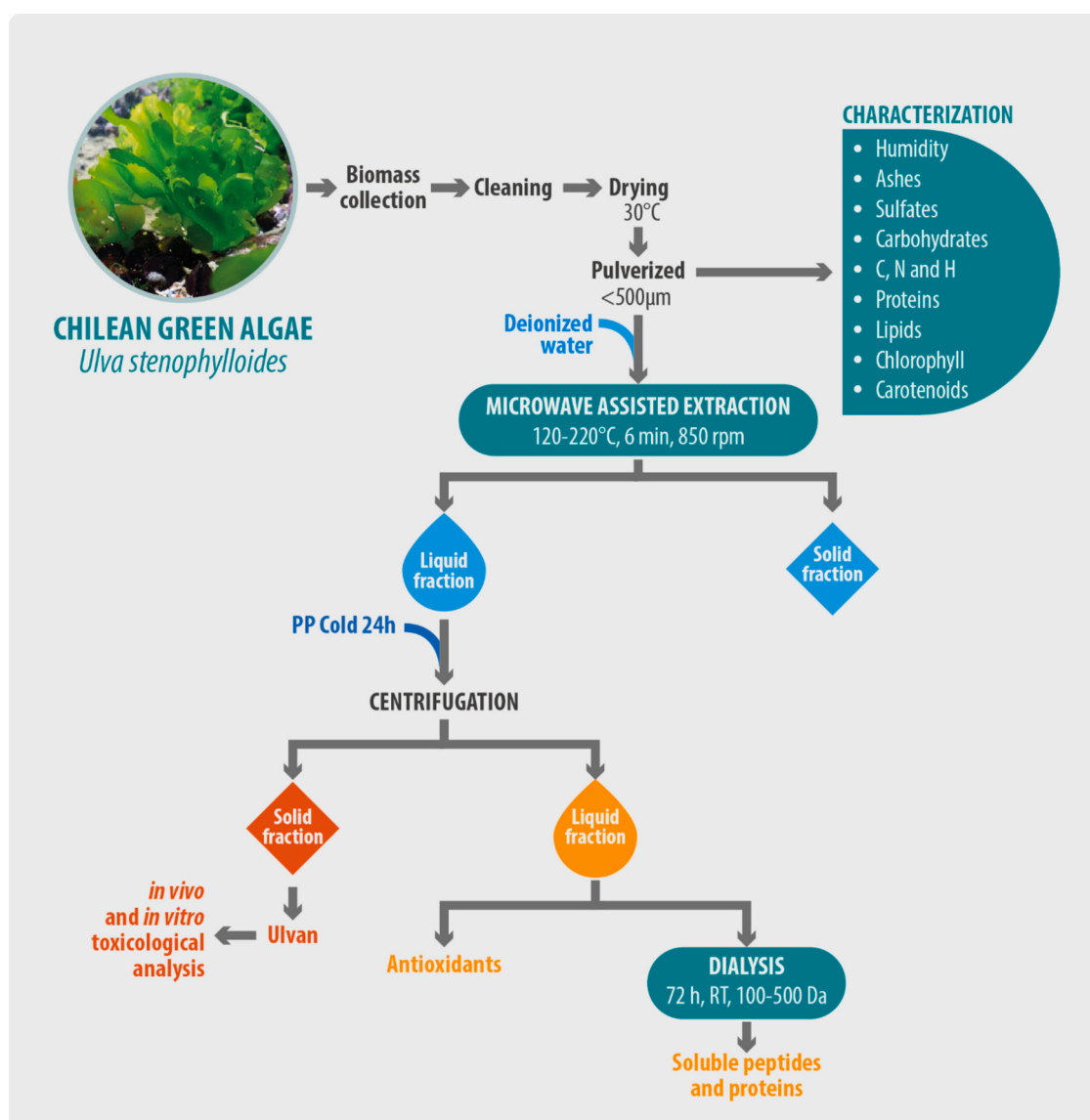


Fig. 1. General diagram of the biorefinery process for extracting biocompounds from *Ulva stenophylloides*.

rank-sum test was carried out with Dunn's post-hoc multiple comparison test to determine statistically significant differences between treatments. For the statistical comparison of MTT measurements performed in N2A cells, one-way ANOVA followed by Dunnett's post-hoc test was performed.

3. Results and discussion

3.1. Characterization of the dry biomass of *U. stenophylloides*

The humidity of the dry biomass utilized for the characterization and biorefinery analyses was 8.6 %. The ash content in this biomass was 20.6 ± 0.3 %, similar to that obtained for *U. lactuca* [50]. Furthermore, these results are consistent with the high ash values identified in green seaweeds [37]. As other green seaweeds, *Ulva* spp. has an interesting composition of minerals, which can be seen as a nutritional advantage, making it a potential ingredient to boost the mineral content of food [51].

The sulfate (–OSO₃H) content was 246 ± 4.3 mg/g (24.6 %), which falls within the range determined in other *Ulva* species [31]. It is important to consider that the sulfate group is a significant fragment in *Ulva* composition, as it is a central moiety in ulvan structure [31,52]. Additionally, it has been determined that sulfated seaweed polysaccharides exhibit antioxidant activities, which are closely related to the –OSO₃H content. For example, [53] found that the scavenging capacity of superoxide anions is reduced with a lower sulfate content in *Laminaria japonica*. Therefore, we can hypothesize that *Ulva stenophylloides* has high antioxidant capacity during ROS bursts.

The use of *Ulva* species has become more prominent due to global occurrences of green tides [54]. *Ulva* oligosaccharides, derived from polysaccharide degradation, demonstrate distinct activities (anti-inflammatory, antioxidant, antiviral, anticancer, immune-regulating, and anti-hyperlipidemic activities, among others), and offer exceptional solubility and bioavailability [55]. In this study, we evaluated the content of oligosaccharides present in the dry biomass of *U. stenophylloides* before and after microwave-assisted extraction treatment. The

Table 1
Total oligosaccharide content of dry *Ulva stenophylloides* biomass extracted pre-biorefinery (GIA: glucuronic acid; Glc: glucose; Xyl: xylose; Rha: rhamnose). The results were expressed as mean value ± SD (n = 3).

Oligosaccharide	Content (mg/g)	%
GIA	38 ± 6	3.8
Glc	81 ± 3	8.1
Xyl	36 ± 2	3.6
Rha	80 ± 4	8.0

Table 2
Chlorophyll (*Chl*) and carotenoid contents in the dry biomass of *U. stenophylloides*. The results were expressed as mean value ± SD (n = 3).

Pigment	Content (mg/g)	%
<i>Chl</i> “a”	154 ± 30.9	15.41
<i>Chl</i> “b”	110 ± 29.6	11.02
Total Carotenoids	46.2 ± 8.9	4.62

Table 3
Yield of solid, liquid (solubilized extract), and ulvan fractions obtained via microwave-assisted extraction from *U. stenophylloides* biomass.

Temperature	Yield (%)		
	Solid fraction	Liquid fraction (solubilized extract)	Ulvan
120 °C	39.02 ± 0.95	26.88 ± 6.20	20.87 ± 3.88
140 °C	34.29 ± 4.92	31.38 ± 7.94	15.03 ± 1.00
160 °C	28.51 ± 0.42	49.68 ± 8.43	10.53 ± 0.61
180 °C	22.59 ± 1.93	63.88 ± 17.87	10.41 ± 3.68
200 °C	15.81 ± 2.73	37.93 ± 10.49	11.06 ± 0.29
220 °C	15.57 ± 0.78	48.83 ± 3.57	12.07 ± 1.82

oligosaccharide content extracted pre-biorefinery is shown in Table 1; the total carbohydrate content of *U. stenophylloides* was approximately 27 %, consisting mainly of glucose [81 mg/g (8.1 %)], rhamnose [80 mg/g (8 %)], and glucuronic acid [38 mg/g (3.8 %)]. These results are higher than the total sugar content described for *U. lactuca* but maintain the trend observed in several *Ulva* species [37,55,56]. It will be interesting to carry out future research on this group of oligosaccharides using this waste biomass along the South Pacific coast.

The C, H, and N contents were 31.5 %, 5.11 %, and 4.11 %, respectively, similar to the concentrations determined in previous studies on *Ulva* species worldwide [14,37,57–60]. The same trend was observed for the total protein concentration (21 %). The lipid content of the biomass was 2.1 ± 0.4 %, a concentration similar to that identified for *Ulva lactuca* and a complex of *Ulva* spp. [50,57,60], comprising 61 % palmitic acid and 39 % linolenic acid, which is an ω-6 fatty acid [61]. It is known that lipids, comprising less than 5 % of seaweed biomass, are essential molecules that can act as signaling compounds to control diverse biological processes, and are a determinant factor of biomass quality [62,63]. Additionally, as observed in diverse species of *Ulva* and other seaweed species, lipid levels could vary among seasons [60], given that seasonal stressful conditions modulate resistance/defense physiological mechanisms [62,64]. In this context, it would be interesting to characterize the lipidomic signature in *Ulva stenophylloides*, considering their presence and coverage in Chilean marine green tides throughout all seasons. Such a study could contribute to the ecophysiological explanation of their high environmental stress tolerance.

Chlorophylls are important antioxidant biomolecules [65,66], which could be part of the tolerance mechanisms of *U. stenophylloides* to stress. Pigment analysis demonstrated variable contents compared to other *Ulva* species (see Table 2). For example, the contents of chlorophyll “a” and “b” were 13 % and 46 % higher, respectively, in *U. stenophylloides* than in *Ulva rigida* [63]. The chlorophyll contents of *U. stenophylloides* (Table 2) fell within the range determined in several *Ulva* species [41,63,65].

Like chlorophylls, carotenoids are important antioxidant biocompounds [66,67], and in *Ulva* spp., they have been identified as key lipidic components of the antioxidant system [67]. The total content of carotenoids was 4.6 % (see Table 2), similar to the values determined in *Ulva rigida* [63] and different *Ulva* spp. from both natural habitats and aquaculture settings [67]. Therefore, we expect that different carotenoid compounds, as another set of antioxidant molecules, will be expressed differentially in *U. stenophylloides* during green tide growth across seasons and geographic locations, among other environmental conditions.

3.2. Analysis of soluble extracts: ulvan, oligosaccharide content, and antioxidant activity

The yield of the solid phases of *U. stenophylloides* was strongly correlated with temperature (correlation coefficient = –0.97), with lower yields observed at higher extraction temperatures (15 %, see Table 3). The yield of the liquid phase (solubilized extract) was weakly correlated with temperature (correlation coefficient = 0.31), with the highest yield recorded at 180 °C (64 %) and the lowest at 120 °C (27 %). Increasing the temperature favors the opening of the cell wall for the extraction of active compounds and, consequently, higher solubilization

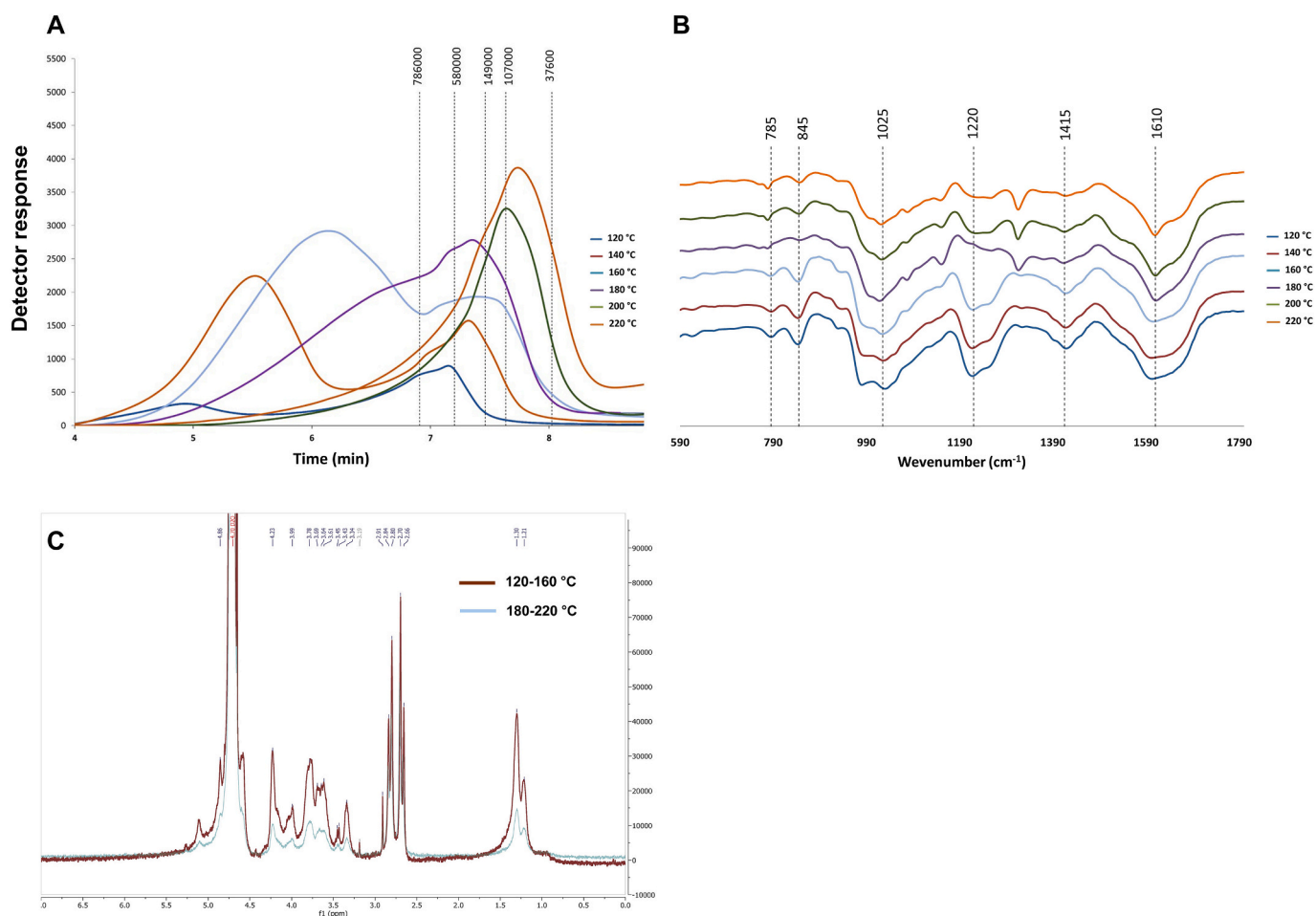


Fig. 2. Structural characteristics of ulvans extracted from *U. stenophylloides*. (A) High-performance size-exclusion chromatography (HPSEC) of ulvans extracted at different temperatures (120–220 °C) via microwave-assisted extraction, and (B) Fourier-transform infrared attenuated total reflectance (FTIR-ATR) at the same temperatures of extraction, and (C) proton nuclear magnetic resonance (¹H NMR).

of extracts in the recovered liquid phase is attained [37]. Finally, the yield of ulvans was also correlated with temperature (correlation coefficient = −0.73), with the highest yield observed at 120 °C (21 %) and

the lowest at 160 °C and 180 °C (10 %) (see Table 3). These results are consistent with previous observations across the *Ulva* genus [31,68], whereby extractions at higher temperatures result in lower yields due to

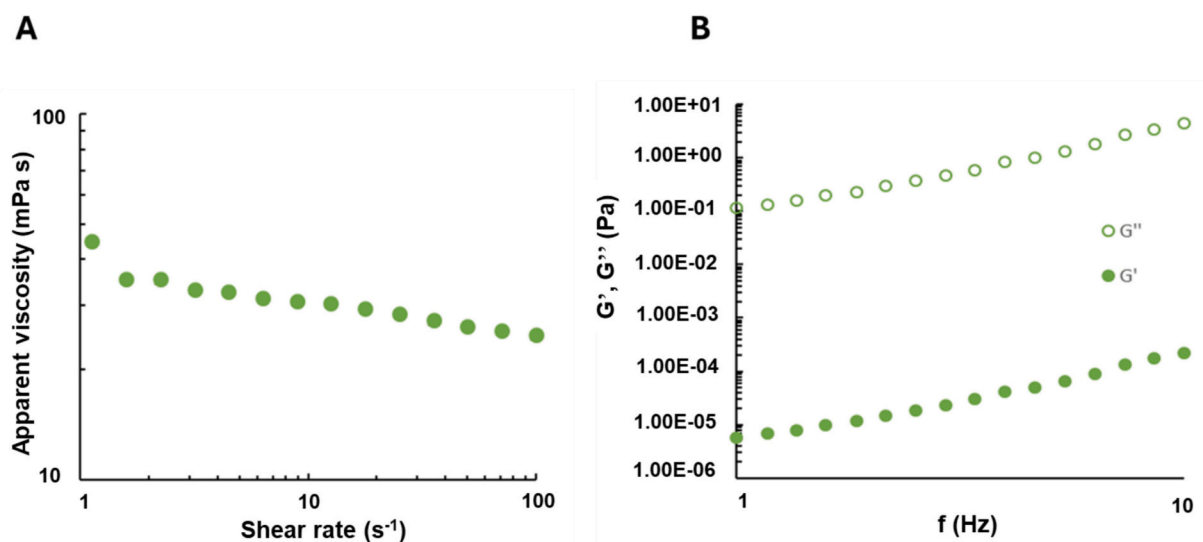


Fig. 3. Rheological properties of aqueous solutions of ulvans (1 %, w/w) measured at 25 °C: (A) steady-state shear and (B) small-amplitude oscillatory shear measurements.

the interaction of ulvans with other cell components [68]. However, in some cases, such as *Ulva meridionalis* and *Ulva ohnoi* [69], no correlation was found between temperature and ulvan yield, with the optimal extraction temperature observed at 160 °C. The microwave method employed in this study effectively improved the efficiency of ulvan yield extraction. Nevertheless, it may be worthwhile to explore extraction at lower temperatures to determine the optimal range for ulvan production from *U. stenophylloides*.

3.2.1. Characterization of ulvan fraction

Fig. 2A shows the high-performance size-exclusion chromatography (HPSEC) profiles of ulvans extracted at different temperatures (120–220 °C) via microwave-assisted extraction. The observed molecular weight distributions exhibit two peaks for ulvans extracted at low temperatures, while a single peak in the region of low molecular weights was identified at high temperatures (160–220 °C) (Fig. 2A). However, the lowest molecular weight observed corresponded to 40–200 kDa, in a range higher than those attained with oxalic acid [70,71] but lower than with enzyme-assisted extraction [72].

Fig. 2B displays the FTIR spectra of polysaccharides extracted under different microwave conditions; in all cases, the typical peaks of ulvans were recorded as previously described by other authors [37,68]. The band identified at around 1025 cm⁻¹ corresponds to the symmetric linkage stretching of C—O—C. The symmetric and asymmetric bands observed at around 1610 cm⁻¹ and 1415 cm⁻¹ are characteristic of the carboxyl group in uronic acid. The band observed at 1220 cm⁻¹ corresponds to the stretching of S=O bond in the sulfate group (Fig. 2B). It is important to mention that this peak almost disappears at 180, 200, and 220 °C, indicating the release of the sulfate group from polysaccharides at higher temperatures. The lowest peak observed at around 845 cm⁻¹ corresponds to the C—O—S stretching vibration of ulvans.

The ¹H NMR spectra of ulvans obtained at 120–160 °C and 180–220 °C are present in Fig. 2C. More intense signals were obtained for ulvans extracted at 120–160 °C. The signals with chemical shifts between 1.20 and 1.30 ppm can be attributed to the methyl proton of rhamnosyl residues, while signals at 2.5–3.0 ppm would be related to the O-acetyl group due to the presence of acetylated polysaccharide [73]. The authors of [74] reported the presence of rhamnose, glucuronic acid, and iduronic acid residues in *U. fasciata* extracts with a spectral pattern very similar to that found in this work. The α-anomeric protons of rhamnose and iduronic acid could be observed at 5.13 ppm, while the β-form at 4.86 ppm overlapped with the solvent signal. The intense signals with chemical shifts between 3.34 and 4.23 ppm were assigned to glucuronic acid.

Fig. 3 presents the rheological properties of ulvans with the highest performance, as representative of the extracted biopolymers. The apparent viscosity decreased with increasing shear rate (Fig. 3A), exhibiting shear-thinning behavior that is characteristic of these types of biopolymers [75]. Previous research reported values around 10 Pa s (1 s⁻¹) and 0.1 Pa s (100 s⁻¹) at 20 °C for similar biopolymer contents extracted from *U. linza* in the presence of CaCl₂ (7 mM). The rheological properties of ulvans are highly dependent not only on the *Ulva* species and collection season [68] but also on the extraction method. The mechanical spectrum (Fig. 3B) displays a characteristic liquid-like behavior with the viscous modulus (G'') higher than the elastic modulus (G') over the tested frequency range. The presence of bio-ionic liquids during the microwave hydrothermal extraction treatment using *Ulva* spp. led to the development of gelled matrices [37]. It was also reported that the elastic features of ulvan gelled matrices from other *Ulva* species were directly proportional to both their molar weight and the quantity of sulfate, glucuronic acid, and rhamnose [37,75].

3.2.2. Antioxidant analysis

Fig. 4 presents the effect of microwave hydrothermal treatment on the antioxidant properties of the soluble extracts of *Ulva stenophylloides*. The content of phenolic compounds demonstrated a notable dependence

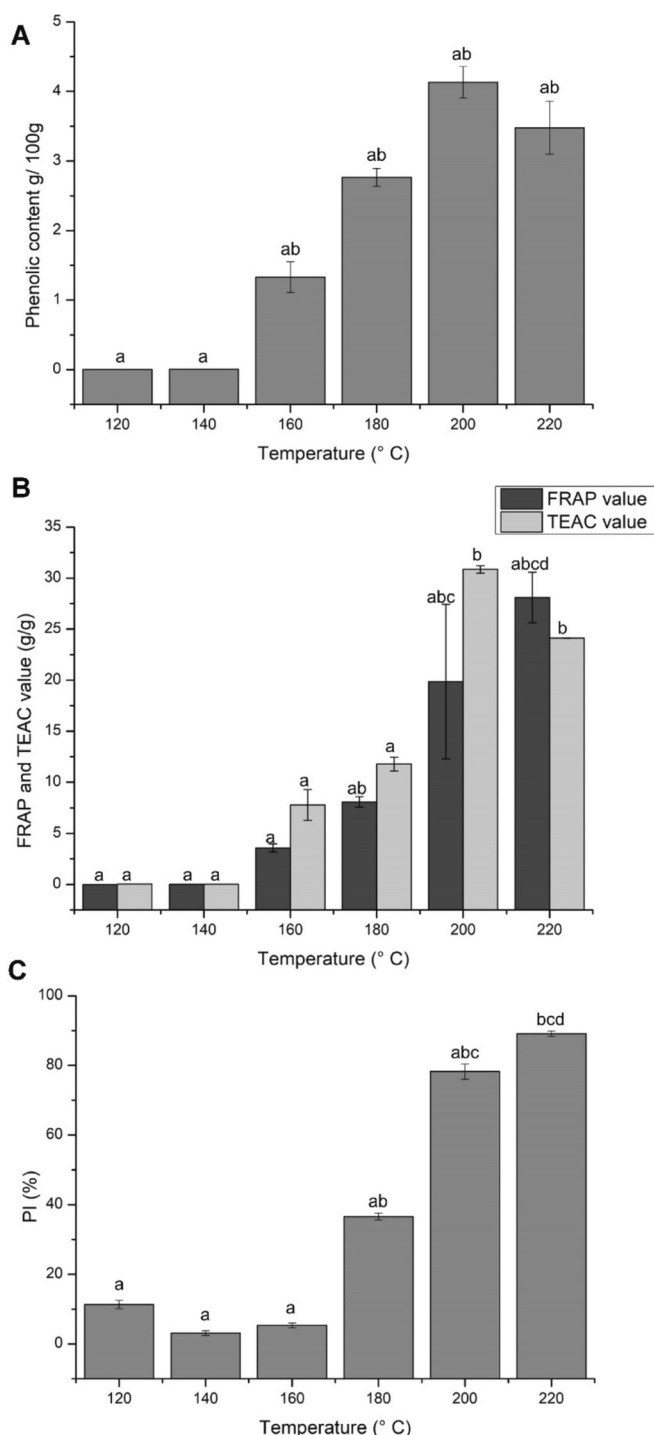


Fig. 4. Influence of microwave-assisted extraction temperature on the antioxidant properties of the liquid phase: (A) phenolic content (g/100 g extract), (B) FRAP and TEAC values (g/g extract), and (C) percentage of inhibition of DPPH. Different letters indicate significant differences according to Dunn's test. The results were expressed as mean value ± SD (n = 3).

on the thermal conditions of the microwave processing (Fig. 4A), with values increasing drastically with temperature. Similarly, the FRAP and TEAC assays and the percentage of inhibition (DPPH assay) showed a similar trend (Fig. 4B–C), with the highest percentage of inhibition (approximately 90 %) obtained in samples extracted at the highest temperatures (200–220 °C) (Fig. 4C). These results suggest that to enhance the antioxidant properties present in *U. stenophylloides*, high

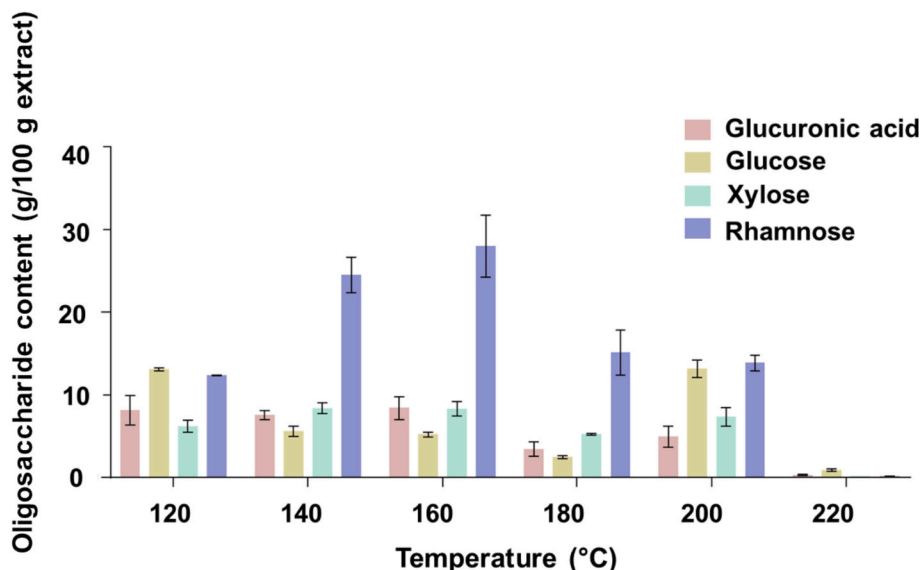


Fig. 5. Influence of microwave-assisted extraction temperature on oligosaccharide content (g/100 g) from the liquid phase of *Ulva stenophylloides* post-hydrolysis. The results were expressed as mean value $\pm$ SD (n = 3).

temperatures should be used in the extraction process, and part of this capacity is related to the high phenolic content.

In this work, we obtained high values of antioxidant properties, following similar trends as observed in some previous studies. However, we obtained higher values than those described for other species of the genus *Ulva* [37,76,77], as well as for some species of brown and red algae, such as *Undaria pinnatifida* [78] and *Sarcopeltis skottsbergii* [79]. Thus, considering the importance of antioxidants for human and animal health [80,81], these results highlight this Chilean green-tide biomass as a relevant sustainable source of antioxidant biomolecules. Of course, further studies are needed to determine all the components associated with this high capacity.

3.2.3. Oligosaccharide content

After hydrolysis, the oligosaccharide extraction from *U. stenophylloides* focused on the extraction of glucuronic acid, glucose, xylose, and rhamnose at temperatures ranging from 120 °C to 220 °C. Fig. 5 demonstrates that the content of oligosaccharides is temperature-dependent, with the highest peaks of glucuronic acid [8.4 g/100 g (8.4 %)], xylose [8.4 g/100 g (8.4 %)], and rhamnose [28 g/100 g (28 %)] observed at 160 °C. These results indicate that an optimal temperature of 160 °C is required for the release and extraction of specific oligosaccharides from *U. stenophylloides*. However, the glucose content showed higher peaks at both 120 °C and 200 °C [13.1 g/100 g (13 %)]. These findings highlight the complexity of the extraction process and the diverse nature of oligosaccharide composition. Furthermore, all the saccharides studied showed their lowest peaks at 220 °C (Fig. 5), with concentrations less than 1 %, suggesting that microwave-assisted extraction is suboptimal at this temperature, resulting in reduced oligosaccharide content.

A comparison with previously reported data on *Ulva* species suggests that an increase in glucose content at higher temperatures (200 °C) promotes starch solubilization, which may be a potential mechanism underlying the observed temperature-dependent variations in oligosaccharide composition in *U. stenophylloides* [37,69,77]. In addition, the rhamnose content in *U. stenophylloides* is remarkably higher than the content of glucose, as described in previous studies on green algae such as *U. lactuca* [59,82].

In addition, [37] reported a decrease in rhamnose, fucose, glucuronic acid, and iduronic acid contents with increasing hydrothermal temperature, while glucose and xylose showed the opposite trend. These results

also emphasize the effect of temperature on oligosaccharide composition and establish that *Ulva* oligosaccharides are stable up to 160 °C, with degradation starting at 180 °C. Therefore, controlling the temperature during the extraction process to maintain the integrity of oligosaccharides is crucial for the sustainable and productive extraction of oligosaccharides from the *Ulva* genus. The results obtained from *U. stenophylloides* suggest that this biomass can be optimally used for the production of oligosaccharides, which are commonly recognized for their high bioactivity [55].

3.3. Toxicological assays of ulvan fraction

3.3.1. In vitro analysis

Comprehensive cytotoxicity studies on ulvans have previously been conducted using various cell lines, including macrophages, gut cells,

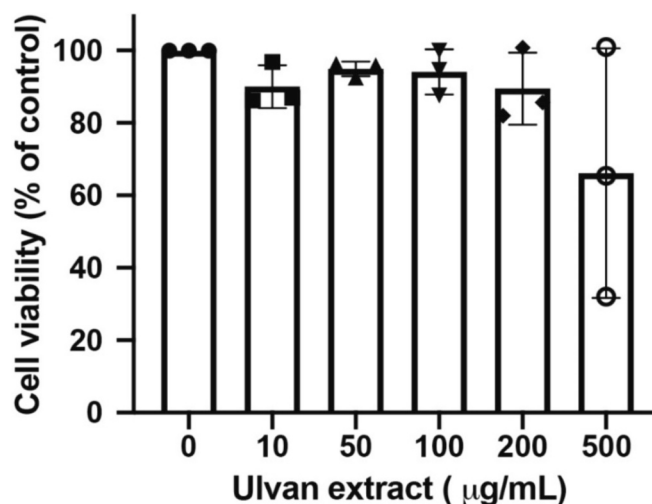


Fig. 6. Non-toxic effects of ulvan extract on N2A cells. N2A cells were seeded on low-serum cultured media and, after 24 h, the cells were treated with different concentrations of ulvan extract (0–500 µg/mL) dissolved in PBS. The results were expressed the mean of three different assays performed in quadruplicate. One-way ANOVA followed by Dunnett's post hoc test was used to analyze differences between the control cells (0) and cells exposed to different ulvan extract concentrations (10–500 µg/mL).

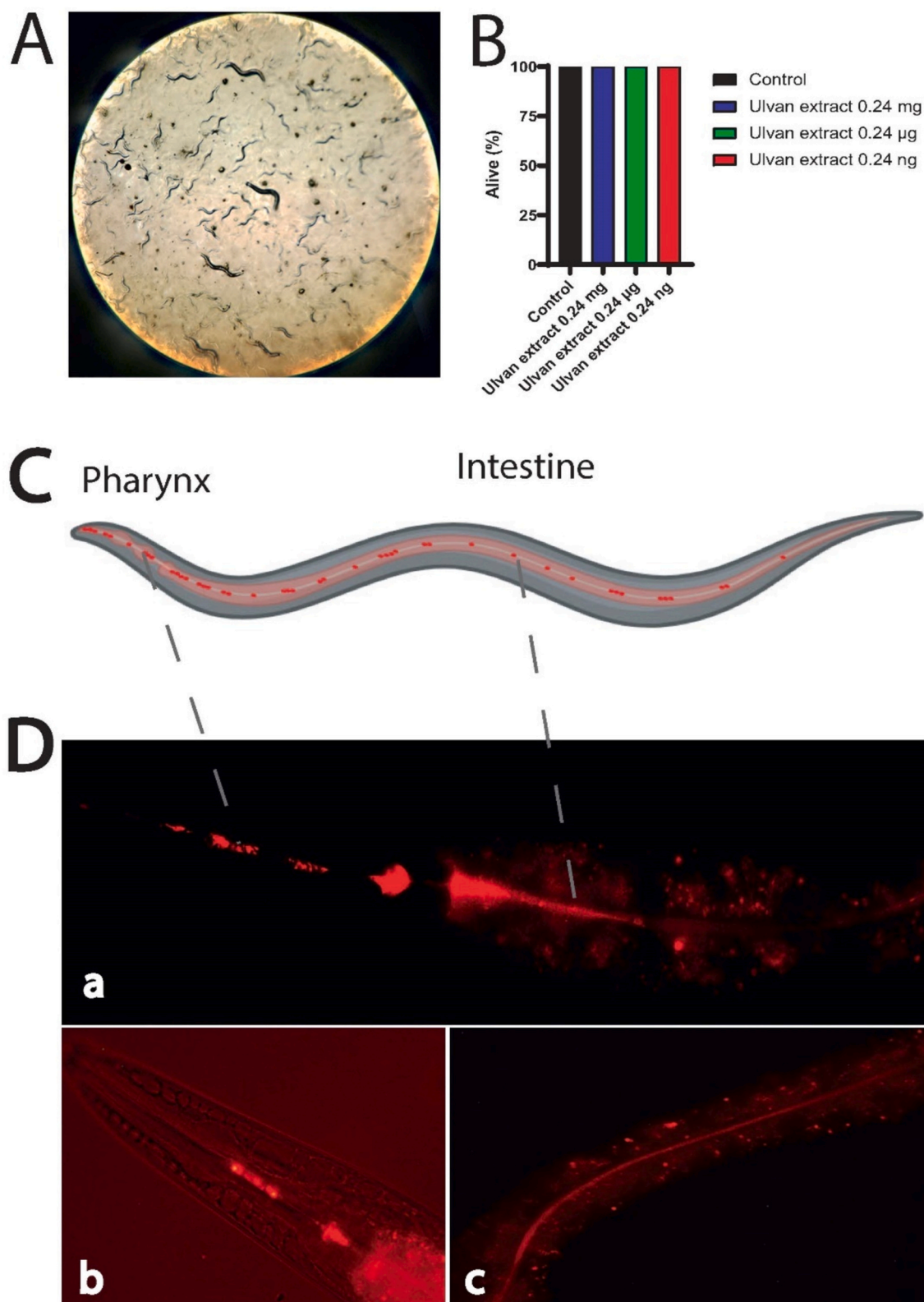


Fig. 7. Non-toxic effects of ulvan extract on *C. elegans* worms. (A) Representative image of *C. elegans* worms cultured on solid agar plates supplemented with the ulvan extract, alongside the bacterial strain OP50. (B) Survival analysis: graph illustrating the percentage of survival of adult worms after 48 h of exposure to the ulvan extract. The results were expressed as the mean $\pm$ SEM of 90 individuals per condition of three independent assays. (C) Diagram showing a *C. elegans* individual fed with the algal extract. (D) Red autofluorescence observed along the nematode body under fluorescence microscope (400 $\times$ magnification). Detailed views of the pharynx (a and b) and intestinal lumen (c) highlighting the presence of the extract. The ulvan extract was dissolved in ethanol and diluted with OP50 before seeding onto cultured plates, following established protocols [94,95]. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

mouse fibroblasts, and carcinogenic cells. The obtained results have been promising, demonstrating that ulvans are largely non-toxic. Notably, ulvan fractions derived from different *Ulva* species showed cell viability rates exceeding 50 % in cultures exposed to doses as high as 500 µg/mL of the sample. These findings underscore the potential of ulvans as a safe biomaterial for diverse medical applications, paving the way for further research into their therapeutic uses [31,83].

N2A cell is a mouse neuroblastoma cell line with conserved signaling pathways and neuronal-like features comparative to human neuronal cell lines and primary culture mouse neurons. It is also widely used to study the cytotoxicity of marine toxins, polysaccharides, and other compounds [84–86]. Therefore, we used N2A cells to study the cytotoxicity of the ulvan extract obtained from *U. stenophylloides*.

Fig. 6 shows the viability results of N2A cells treated with different concentrations of the ulvan extract (10–500 µg/mL) using the MTT assay. We found that the ulvan extract precipitated when low-serum media were used to treat N2A cells (data not shown); therefore, we diluted the extract in PBS up to a concentration of 1000 µg/mL stock solution. For each point, a control was performed containing just PBS since PBS diluted the serum-supplemented media. We found that the ulvan extract had no toxic effect up to a concentration of 500 µg/mL (Fig. 6). The results of this last point present more variability than the other points due to the reduced concentration of serum obtained when mixing PBS with the cell culture media. These results demonstrate that, as found in other cell lines with different ulvan extracts from different species [31], the ulvan extract from *U. stenophylloides* is largely non-cytotoxic *in vitro*.

3.3.2. *In vivo* analysis

We expanded our *in vitro* studies by examining the cytotoxicity of the ulvan extract from *U. stenophylloides* *in vivo*. We used the nematode *Caenorhabditis elegans* as our model organism, as this species is widely used in toxicity assays due to its ability to provide whole-organism cytotoxicity data. This model helps understand the potential effects of an extract on a complete biological system, serving as a bridge between cellular studies and higher-order organisms [87]. Fig. 7 shows that supplementation with the ulvan extract has non-toxic effects on *C. elegans* worms. Adult worms of one day old were fed with different concentration of the extract for 48 h. Fig. 7A presents a representative image displaying the various developmental stages of worms cultured on solid agar plates supplemented with 2.4 mg/mL ulvan extract. The graph depicts the survival rate of adult worms after 48 h of exposure to varying concentrations of ulvan extract. All tested concentrations showed 100 % survival (Fig. 7B). Observation of individuals under fluorescence microscope revealed the incorporation of the ulvan extract by these worms, as evidenced by the presence of ulvan autofluorescence in the pharynx and intestinal lumen (Fig. 7C–D). It is remarkable that the administration of the ulvan extract to *C. elegans* does not result in any adverse toxicological effects, as evidenced by the survival of the worms and their successful incorporation of the extract. These findings could be the starting point for further studies on the biological potential of these algae, wherein *C. elegans* has been positioned as an animal model to determine molecular pathways of action and potential applications in biomedicine [88–90]. Additionally, previous studies have shown that a variety of polysaccharides derived from algae and fungi confer a range of biological benefits in *C. elegans*. For instance, fucoidan has been shown to improve health span following *Helicobacter pylori* infections [91], while lentinan enhances lifespan and oxidative stress resistance [92]. Moreover, ulvan has been found to reduce alpha-synuclein fibrillogenesis [93]. The incorporation of algal extract in *C. elegans* has been reported previously [94], and this extract has demonstrating various physiological benefits [e.g., Patentscope WO/2020/257952 (anti-proliferative effect in prostate cancer), WO/2021/097584 (preventing health conditions that require neuroprotection)]. Collectively, these studies highlight the potential of marine polysaccharides in enhancing health and longevity, suggesting that ulvans from *U. stenophylloides*

could be further evaluated for their benefits at both the animal and human levels.

4. Conclusion

In this study, we developed an efficient biorefinery process for the extraction of various compounds from the underutilized South Pacific green-tide species *Ulva stenophylloides*. Our study found that temperatures between 120 and 220 °C in the microwave treatment affect the yields of the solid, liquid, and ulvan fractions, as well as the recovery of biocompounds from the biomass. This highlights the importance of maintaining an optimal temperature for each bioactive fraction during extraction. Additionally, the ulvan extracts were found to be non-toxic in both *in vitro* and *in vivo* analyses. Thus, our research underscores the effective use of all biomass components and emphasizes the identification and utilization of fractions with potential pharmaceutical and cosmetic applications due to their high added value.

CRedit authorship contribution statement

Loretto Contreras-Porcia: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **María Dolores Torres:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Herminia Domínguez:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Julie Queffelec:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Formal analysis. **Florentina Piña:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Formal analysis. **Sheyma Inoubli:** Writing – review & editing, Writing – original draft, Methodology, Investigation. **Rebeca Aldunate:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation. **Francisca C. Bronfman:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation. **Nancy Pizarro:** Writing – review & editing, Writing – original draft, Methodology, Investigation. **Fernanda Toro-Mellado:** Writing – review & editing, Writing – original draft, Methodology, Investigation. **Maximilian Pozo-Sotelo:** Writing – review & editing, Writing – original draft, Methodology, Investigation. **María Gallardo-Campos:** Writing – review & editing, Writing – original draft, Methodology, Investigation. **Benjamín Pinilla-Rojas:** Writing – review & editing, Writing – original draft, Methodology, Investigation. **Jorge Rivas:** Writing – review & editing, Writing – original draft, Methodology, Investigation. **Jean Pierre Remonsellez:** Writing – review & editing, Writing – original draft, Methodology, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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